Notomath—LaTeX math support for the noto package

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March 22, 2021

This package provides math support for the Google font collection Noto, a massive text font whose LaTeX support has been available for several years using Bob Tennent's noto package. The math support is based on newtxmath but there are some wrinkles that make it desirable to craft a small package, notomath, that can serve as a front end to simplify the business of lining up the text and math options, given that there are considerable size discrepancies between text and math at their natural sizes.

For the noto option to newtxmath, the Roman and Greek alphabets in the latter were substituted by those in Noto scaled down by 10% to an x-height of 482, which is a close enough match to the symbols in newtxmath for all practical purposes.

The Noto fonts comprise three different faces: NotoSerif, NotoSans and NotoSansMono. Each face has its own .sty file: noto-serif.sty, noto-sans.sty and noto-mono.sty. There is also an integrated .sty file, noto.sty, though it is a bit less configurable. Most of the time, it should not be necessary to load any of these packages explicitly, that task being relegated to the package notomath.

Usage

For most users, it will likely suffice to place some small variant of the following line in the preamble:

```
\usepackage{notomath}
```

The effect of this is:

- load noto-serif and noto-sans scaled down by the factor .9 to an x-height of 482;
- set the main text font to NotoSerif and set \sfdefault to NotoSans;
- the only weight used from the nine available weights are regular and bold, as these are the weights used in newtxmath with options noto and notosans;
- load newtxmath with option noto at natural scale.

Alternatively, the package may be loaded with options that modify the above behaviors:

- mono loads, in addition, noto-mono at the same x-height as the other Noto text packages.
- scale (or scaled) allows you to rescale all the Noto text packages and newtxmath by the specified factor.
- the figure style for NotoSerif and NotoSans may be controlled by the options proportional (or pf) and oldstyle (or osf), as in the noto package. (The default setting is tabular, lining figures.)
- You may add as an option to notomath any newtxmath option that is relevant to noto—these are simply passed along to newtxmath, if truly relevant. (E.g., option vvarbb would be passed along, but not garamondx because that would change all the math italic alphabets to match garamondx.)
- sfdefault changes the main text font to NotoSans, but leaves the meaning of \rmdefault unchanged, so that \textrm{} prints its argument using NotoSerif.

Usage Notes

- There are a couple of issues that might lead you to avoid NotoSansMono as your Typewriter font:
  - In OT1 encoding, the glyphs are not laid out as TEX TYPEWRITER TEXT, as, for example, cmtt. This means you will get incorrect output from text that involves quotes, backslash, braces and the like. (This is not a problem in other encodings such as T1.)
  - The NotoSansMono fonts have no visiblespace glyph, so \verb* will fail to to render the space as something like ␣. If this is important to you, replace noto-mono with a package like inconsolata, if you want to try another sans mono font. The loading order is important—you should load inconsolata before loading notomath.
- If you chose the sfdefault option so that NotoSans is the main text font, you may find that a SansMono font if too similar to be easily distinguished from the main font, in which case you may wish to switch to a serifed mono font. If not loading NotoSansMono by means of the option mono, you should load a replacement TT package BEFORE loading notomath if you wish to be able to use the macro \mathtt using glyphs that match those used for \texttt. There are three reasonable options, and possibly more that I'm not aware of. Each would need to be scaled up a bit.
  - The TT package zlmtt does have a visiblespace glyph and its OT1 encoding is in TEX TYPEWRITER TEXT so \verb and \texttt function correctly even in OT1 encoding. I find the caps too tall to be a very good match.
  - The TT package newtxtt does have a visiblespace glyph and works well in T1 encoding. There is no OT1 encoded version currently. Caps are a bit too tall to be a very good match.
  - The TT package nimbusmononarrow does have a visiblespace glyph and its OT1 encoding is in TEX TYPEWRITER TEXT so \verb and \texttt function correctly even in OT1 encoding. Caps are not too tall—this is my preferred serifed TT with NotoSans text.
-

Example preamble fragments

EXAMPLE 1:

```
\usepackage[mono,vvarbb,upint]{notomath}
% load NotoSerif, NotoSans, NotoSansMono, mainfont=NotoSerif
% options vvarbb and upint passed to newtxmath, resulting in
% STIX Blackboard Bold and upright integrals rather than slanted
```

The Noto fonts will be scaled to x-height 482, matching math symbols. The main text font will be NotoSerif.

EXAMPLE 2:

```
\usepackage[varq,varl]{inconsolata} % inconsolata sans mono
\usepackage[vvarbb,uprightscript]{notomath}
% load NotoSerif, NotoSans, mainfont=NotoSerif
% options vvarbb and uprightscript passed to newtxmath, resulting in
% STIX Blackboard Bold and upright script
```

The Noto fonts will be scaled to x-height 482, matching math symbols. The main text font will be NotoSerif.

EXAMPLE 3:

```
\usepackage[scaled=1.12]{nimbusmononarrow}% typewriter font
\usepackage[sfdefault,subscriptcorrection]{notomath}
% load NotoSerif, NotoSans, mainfont=NotoSans
% option subscriptcorrection passed to newtxmath
```

will output the Noto fonts scaled to x-height 482 with matching math symbols. The main text font will be NotoSans.

EXAMPLE 4:

```
\usepackage[scaled=1.24]{nimbusmononarrow}% typewriter font
\usepackage[scale=1.11,sfdefault,pf,osf]{notomath}
% load NotoSerif, NotoSans, mainfont=NotoSans
% option subscriptcorrection passed to newtxmath
```

will output the Noto fonts scaled to x-height 536 with matching math symbols. The main text font will be NotoSans with proportional oldstyle figures except in math, which always uses tabular lining figures.

The examples above all work with `pdflatex`, and with `xelatex` if some additional rules are followed. With `xelatex`, the lines in the above examples must precede the loading of `fontspec`, which must use the option `nomath`. After that, one may load any text fonts required for secondary use, or even replace the main Noto fonts.

Subscript Correction

The spacing of math letters was adjusted so the superscripts would not collide with the base letters. This was necessary mainly for letters like *j*, *f*, *y* and *β* as superscripts and like *D* and Ω as base letters. As a result of these adjustments, some of the formerly problematic superscript letters become problematic subscript letters. Two files are provided to make adjustments to the letter by inserting appropriate kerns when that letter is the first character in a subscript—one for NotoSerif and one for NotoSans letters, under the respective names

```
newtx-noto-subs.tex % for NotoSerif
newtx-notosans-subs.tex
```

The appropriate file is read in by `newtxmath` provided you add the option `subscriptcorrection`. A line in the file of the form `{j}{-2}` will translate to a kern of `-2mu` being placed before a leading *j* in a subscript.

Lower level settings

It may be that you wish to make use of lower level settings in the individual `noto-` packages. In that case, the following information may be useful.

Recall from the README to the `noto` package:

- `\usepackage{noto}`
  - loads NotoSerif as `\rmdefault`;
  - loads NotoSans as `\sfdefault`;
  - loads NotoSansMono as `\ttdefault`;
  - lets `\familydefault` to `\rmdefault`;
  - so, main body text is NotoSerif, `\textsf` points to NotoSans and `\texttt` to NotoSansMono.
  - The `scale` option does not affect NotoSerif size.
- `\usepackage{noto-serif}`
  - loads NotoSerif as `\rmdefault`;
  - lets `\familydefault` to `\rmdefault`;
  - neither NotoSans nor NotoSansMono is loaded.
  - `scale` option available.

- `\usepackage{noto-sans}`
  - loads NotoSans as `\sfdefault`;
  - does not modify `\familydefault`;
  - neither NotoSerif or NotoSansMono is loaded.
  - `scale` option available.
- `\usepackage[sfdefault]{noto-sans}`
  - loads NotoSans as `\sfdefault`;
  - lets `\familydefault` to `\sfdefault`
  - neither NotoSerif or NotoSansMono is loaded and NotoSans becomes the main text font.
  - `scale` option available.
- `\usepackage{noto-mono}`
  - loads NotoSansMono as `\ttdefault`;
  - neither NotoSerif or NotoSans is loaded.
  - `scale` option available.

At its lowest level, you invoke NotoMath in `newtxmath` using the option `noto`, and NotoSansMath using the option `notosans`.

Math samples

An inversion formula: Let $g : \mathbb{R}^+ \to \mathbb{R}$ be bounded and right continuous, and let $\varphi(\alpha) \coloneqq \int_0^\infty e^{-\alpha t} g(t)\, dt$ denote its Laplace transform. Then, for every $t > 0$,

$$g(t) = \lim_{\varepsilon \to 0} \lim_{\lambda \to \infty} \varepsilon^{-1} \sum_{\lambda t < k \leq (\lambda+\varepsilon)t} \frac{(-1)^k}{k!} \lambda^k \varphi^{(k)}(\lambda). \tag{1}$$

Solutions of systems of ODEs: Let $\mathbf{v}(\mathbf{x}, \boldsymbol{\alpha})$ denote a parametrized vector field ($\mathbf{x} \in U$, $\boldsymbol{\alpha} \in A$) where U is a domain in $\mathbb{R}^n$ and the parameter space A is a domain in $\mathbb{R}^m$. We assume that $\mathbf{v}$ is C^k-differentiable as a function of $(\mathbf{x}, \boldsymbol{\alpha})$, where $k \geq 2$. Consider a system of differential equations in U:

$$\mathbf{x} = \mathbf{v}(\mathbf{x}, \boldsymbol{\alpha}), \qquad \mathbf{x} \in U \tag{2}$$

Fix an initial point $\mathbf{p}_0$ in the interior of U, and assume $\mathbf{v}(\mathbf{p}_0, \boldsymbol{\alpha}_0) \neq \mathbf{0}$. Then, for sufficiently small t, $|\mathbf{p} - \mathbf{p}_0|$ and $|\boldsymbol{\alpha} - \boldsymbol{\alpha}_0|$, the system (2) has a unique solution $\mathbf{x}_{\boldsymbol{\alpha}}(t)$ satisfying the initial condition $\mathbf{x}_{\boldsymbol{\alpha}}(0) = \mathbf{p}$, and that solution depends differentiably (of class C^k) on t, $\mathbf{p}$ and $\boldsymbol{\alpha}$.

Stirling's formula:

$$\Gamma(z) \sim e^{-z} z^{z-1/2} \sqrt{2\pi} \left[1 + \frac{1}{12z} + \frac{1}{288z^2} - \frac{139}{51840z^3} + \dots \right], \quad z \to \infty \text{ in } |\arg z| < \pi. \tag{3}$$

Bézier curves: Given z_1, z_2, z_3, z_4 in $\mathbb{C}$, define the Bézier curve with control points z_1, z_2, z_3, z_4 by

$$z(t) \coloneqq (1-t)^3 z_1 + 3(1-t)^2 t z_2 + 3(1-t) t^2 z_3 + t^3 z_4, \qquad 0 \leq t \leq 1.$$

Because $(1-t)^3 + 3(1-t)^2 t + 3(1-t)t^2 + t^3 = (1-t+t)^3 = 1$ and all summands are positive for $0 \leq t \leq 1$, $z(t)$ is a convex combination of the four points z_k, hence the curve defined by $z(t)$ lies in their convex hull. As t varies from 0 to 1, the curve moves from z_1 to z_4 with initial direction $z_2 - z_1$ and final direction $z_4 - z_3$.

Maxwell's equations:

$$\mathbf{B}' = -c\nabla \times \mathbf{E}$$
$$\mathbf{E}' = c\nabla \times \mathbf{B} - 4\pi\mathbf{J}.$$

Residue theorem: Let f be analytic in the region G except for the isolated singularities $a_1, a_2, \ldots, a_m$. If γ is a closed rectifiable curve in G which does not pass through any of the points a_k and if $\gamma \approx 0$ in G, then

$$\frac{1}{2\pi i}\int_\gamma f = \sum_{k=1}^{m} n(\gamma; a_k)\,\mathrm{Res}(f; a_k).$$

Maximum modulus principle: Let G be a bounded open set in $\mathbb{C}$ and suppose that f is a continuous function on $\bar{G}$ which is analytic in G. Then

$$\max\{|f(z)| : z \in \bar{G}\} = \max\{|f(z)| : z \in \partial G\}.$$

Jacobi's identity: Define the *theta function* ϑ by

$$\vartheta(t) = \sum_{n=-\infty}^{\infty} \exp(-\pi n^2 t), \qquad t > 0.$$

Then

$$\vartheta(t) = t^{-1/2}\vartheta(1/t).$$

The following three samples show the previous three reworked using NotoSerif and its associated math fonts.

Residue theorem: Let f be analytic in the region G except for the isolated singularities $a_1, a_2, \ldots, a_m$. If γ is a closed rectifiable curve in G which does not pass through any of the points a_k and if $\gamma \approx 0$ in G, then

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Font Tables

Sans Math Letters

	′0	′1	′2	′3	′4	′5	′6	′7	
′00x	Γ_{0}	Δ_{1}	Θ_{2}	Λ_{3}	Ξ_{4}	Π_{5}	Σ_{6}	Υ_{7}	″0x
′01x	Φ_{8}	Ψ_{9}	Ω_{10}	α_{11}	β_{12}	γ_{13}	δ_{14}	ϵ_{15}	
′02x	ζ_{16}	η_{17}	θ_{18}	ι_{19}	κ_{20}	λ_{21}	μ_{22}	ν_{23}	″1x
′03x	ξ_{24}	π_{25}	ρ_{26}	σ_{27}	τ_{28}	υ_{29}	ϕ_{30}	χ_{31}	
′04x	ψ_{32}	ω_{33}	ε_{34}	ϑ_{35}	ϖ_{36}	ϱ_{37}	ς_{38}	φ_{39}	″2x
′05x	$\leftharpoonup_{40}$	$\leftharpoondown_{41}$	$\rightharpoonup_{42}$	$\rightharpoondown_{43}$	$\triangleleft_{44}$	$\triangleright_{45}$	$\triangleright_{46}$	$\triangleleft_{47}$	
′06x	0_{48}	1_{49}	2_{50}	3_{51}	4_{52}	5_{53}	6_{54}	7_{55}	″3x
′07x	8_{56}	9_{57}	$.{}_{58}$	$,{}_{59}$	$<_{60}$	$/_{61}$	$>_{62}$	$\star_{63}$	
′10x	∂_{64}	A_{65}	B_{66}	C_{67}	D_{68}	E_{69}	F_{70}	G_{71}	″4x
′11x	H_{72}	I_{73}	J_{74}	K_{75}	L_{76}	M_{77}	N_{78}	O_{79}	
′12x	P_{80}	Q_{81}	R_{82}	S_{83}	T_{84}	U_{85}	V_{86}	W_{87}	″5x
′13x	X_{88}	Y_{89}	Z_{90}	$\flat_{91}$	$\natural_{92}$	$\sharp_{93}$	$\smile_{94}$	$\frown_{95}$	
′14x	ℓ_{96}	a_{97}	b_{98}	c_{99}	d_{100}	e_{101}	f_{102}	g_{103}	″6x
′15x	h_{104}	i_{105}	j_{106}	k_{107}	l_{108}	m_{109}	n_{110}	o_{111}	
′16x	p_{112}	q_{113}	r_{114}	s_{115}	t_{116}	u_{117}	v_{118}	w_{119}	″7x
′17x	x_{120}	y_{121}	z_{122}	$\imath_{123}$	$\jmath_{124}$	$\wp_{125}$	$\vec{}_{126}$	$\hat{}_{127}$	
′20x	$_{128}$	$\varkappa_{129}$	$_{130}$	$_{131}$	0_{132}	1_{133}	2_{134}	3_{135}	″8x
′21x	4_{136}	5_{137}	6_{138}	7_{139}	8_{140}	9_{141}	$\mathcal{A}_{142}$	$\mathcal{B}_{143}$	
′22x	$\mathcal{C}_{144}$	$\mathcal{D}_{145}$	$\mathcal{E}_{146}$	$\mathcal{F}_{147}$	$\mathcal{G}_{148}$	$\mathcal{H}_{149}$	$\mathcal{I}_{150}$	$\mathcal{J}_{151}$	″9x
′23x	$\mathcal{K}_{152}$	$\mathcal{L}_{153}$	$\mathcal{M}_{154}$	$\mathcal{N}_{155}$	$\mathcal{O}_{156}$	$\mathcal{P}_{157}$	$\mathcal{Q}_{158}$	$\mathcal{R}_{159}$	
′24x	$\mathcal{S}_{160}$	$\mathcal{T}_{161}$	$\mathcal{U}_{162}$	$\mathcal{V}_{163}$	$\mathcal{W}_{164}$	$\mathcal{X}_{165}$	$\mathcal{Y}_{166}$	$\mathcal{Z}_{167}$	″Ax
′25x	$\mathcal{a}_{168}$	$\mathcal{b}_{169}$	$\mathcal{c}_{170}$	$\mathcal{d}_{171}$	$\mathcal{e}_{172}$	$\mathcal{f}_{173}$	$\mathcal{g}_{174}$	$\mathcal{h}_{175}$	
′26x	$\mathcal{i}_{176}$	$\mathcal{j}_{177}$	$\mathcal{k}_{178}$	$\mathcal{l}_{179}$	$\mathcal{m}_{180}$	$\mathcal{n}_{181}$	$\mathcal{o}_{182}$	$\mathcal{p}_{183}$	″Bx
′27x	$\mathcal{q}_{184}$	$\mathcal{r}_{185}$	$\mathcal{s}_{186}$	$\mathcal{t}_{187}$	$\mathcal{u}_{188}$	$\mathcal{v}_{189}$	$\mathcal{w}_{190}$	$\mathcal{x}_{191}$	
′30x	$\mathcal{y}_{192}$	$\mathcal{z}_{193}$	$\mathcal{A}_{194}$	$\mathcal{B}_{195}$	$\mathcal{C}_{196}$	$\mathcal{D}_{197}$	$\mathcal{E}_{198}$	$\mathcal{F}_{199}$	″Cx
′31x	$\mathcal{G}_{200}$	$\mathcal{H}_{201}$	$\mathcal{I}_{202}$	$\mathcal{J}_{203}$	$\mathcal{K}_{204}$	$\mathcal{L}_{205}$	$\mathcal{M}_{206}$	$\mathcal{N}_{207}$	
′32x	$\mathcal{O}_{208}$	$\mathcal{P}_{209}$	$\mathcal{Q}_{210}$	$\mathcal{R}_{211}$	$\mathcal{S}_{212}$	$\mathcal{T}_{213}$	$\mathcal{U}_{214}$	$\mathcal{V}_{215}$	″Dx
′33x	$\mathcal{W}_{216}$	$\mathcal{X}_{217}$	$\mathcal{Y}_{218}$	$\mathcal{Z}_{219}$	$\mathcal{a}_{220}$	$\mathcal{b}_{221}$	$\mathcal{c}_{222}$	$\mathcal{d}_{223}$	
′34x	$\mathcal{e}_{224}$	$\mathcal{f}_{225}$	$\mathcal{g}_{226}$	$\mathcal{h}_{227}$	$\mathcal{i}_{228}$	$\mathcal{j}_{229}$	$\mathcal{k}_{230}$	$\mathcal{l}_{231}$	″Ex
′35x	$\mathcal{m}_{232}$	$\mathcal{n}_{233}$	$\mathcal{o}_{234}$	$\mathcal{p}_{235}$	$\mathcal{q}_{236}$	$\mathcal{r}_{237}$	$\mathcal{s}_{238}$	$\mathcal{t}_{239}$	
′36x	$\mathcal{u}_{240}$	$\mathcal{v}_{241}$	$\mathcal{w}_{242}$	$\mathcal{x}_{243}$	$\mathcal{y}_{244}$	$\mathcal{z}_{245}$	$_{246}$	$_{247}$	″Fx
′37x	$_{248}$	$_{249}$	$_{250}$	$_{251}$	$_{252}$	$_{253}$	$_{254}$	$_{255}$	
	″8	″9	″A	″B	″C	″D	″E	″F	

Sans Math LettersA

	′0	′1	′2	′3	′4	′5	′6	′7	
′00x	Γ0	Δ1	Θ2	Λ3	Ξ4	Π5	Σ6	Υ7	″0x
′01x	Φ8	Ψ9	Ω10	α11	β12	γ13	δ14	ϵ15	
′02x	ζ16	η17	θ18	ι19	κ20	λ21	μ22	ν23	″1x
′03x	ξ24	π25	ρ26	σ27	τ28	υ29	ϕ30	χ31	
′04x	ψ32	ω33	ε34	ϑ 35	ϖ36	ϱ37	ς38	φ39	″2x
′05x	40	41	42	𝟘43	𝟙44	𝟚45	𝟛46	𝟜47	
′06x	𝟝48	𝟞49	𝟟50	𝟠51	𝟡52	53	∉54	∌55	″3x
′07x	∉56	∌57	:=58	=:59	≠60	=61	{62	}63	
′10x	∂64	𝔄65	𝔅66	ℭ67	𝔇68	𝔈69	𝔉70	𝔊71	″4x
′11x	ℌ72	ℑ73	𝔍74	ℜ75	𝔏76	𝔐77	𝔑78	𝔒79	
′12x	𝔓80	𝔔81	ℜ82	𝔖83	𝔗84	𝔘85	𝔙86	𝔚87	″5x
′13x	𝔛88	𝔜89	ℨ90	ℏ91	ℎ92	𝔞93	𝔷94	Å95	
′14x	𝔈96	𝔞97	𝔟98	𝔠99	𝔡100	𝔢101	𝔣102	𝔤103	″6x
′15x	𝔥104	𝔦105	𝔧106	𝔨107	𝔩108	𝔪109	𝔫110	𝔬111	
′16x	𝔭112	𝔮113	𝔯114	𝔰115	𝔱116	𝔲117	𝔳118	𝔴119	″7x
′17x	𝔵120	𝔶121	𝔷122	𝔱123	𝔍124	𝕀125	𝕁126	̂127	
′20x	128	𝔫129	130	131	𝔸132	𝔹133	ℂ134	𝔻135	″8x
′21x	𝔼136	𝔽137	𝔾138	ℍ139	𝕀140	𝕁141	𝕂142	𝕃143	
′22x	𝕄144	ℕ145	𝕆146	ℙ147	ℚ148	ℝ149	𝕊150	𝕋151	″9x
′23x	𝕌152	𝕍153	𝕎154	𝕏155	𝕐156	ℤ157	𝔞158	𝔟159	
′24x	𝔠160	𝔡161	𝔢162	𝔣163	𝔤164	𝔥165	𝔦166	𝔧167	″Ax
′25x	𝔨168	𝔩169	𝔪170	𝔫171	𝔬172	𝔭173	𝔮174	𝔯175	
′26x	𝔰176	𝔱177	𝔲178	𝔳179	𝔴180	𝔵181	𝔶182	𝔷183	″Bx
′27x	𝕚184	𝕛185	𝔱186	𝕛187	𝑔188	𝑦189	190	191	
′30x	192	𝔸193	𝔹194	ℂ195	𝔻196	𝔼197	𝔽198	𝔾199	″Cx
′31x	ℍ200	𝕀201	𝕁202	𝕂203	𝕃204	𝕄205	ℕ206	𝕆207	
′32x	ℙ208	ℚ209	ℝ210	𝕊211	𝕋212	𝕌213	𝕍214	𝕎215	″Dx
′33x	𝕏216	𝕐217	ℤ218	𝛤219	ℿ220	𝛶221	𝕞222	223	
′34x	224	𝔞225	𝔟226	𝔠227	𝔡228	𝔢229	𝔣230	𝔤231	″Ex
′35x	𝔥232	𝔦233	𝔧234	𝔨235	𝔩236	𝔪237	𝔫238	𝔬239	
′36x	𝔭240	𝔮241	𝔯242	𝔰243	𝔱244	𝔲245	𝔳246	𝔴247	″Fx
′37x	𝔵248	𝔶249	𝔷250	𝛫251	252	253	254	255	
	″8	″9	″A	″B	″C	″D	″E	″F	

Serif Math Letters

	′0	′1	′2	′3	′4	′5	′6	′7	
′00x	Γ_{0}	Δ_{1}	Θ_{2}	Λ_{3}	Ξ_{4}	Π_{5}	Σ_{6}	Υ_{7}	″0x
′01x	Φ_{8}	Ψ_{9}	Ω_{10}	α_{11}	β_{12}	γ_{13}	δ_{14}	ϵ_{15}	
′02x	ζ_{16}	η_{17}	θ_{18}	ι_{19}	κ_{20}	λ_{21}	μ_{22}	ν_{23}	″1x
′03x	ξ_{24}	π_{25}	ρ_{26}	σ_{27}	τ_{28}	υ_{29}	ϕ_{30}	χ_{31}	
′04x	ψ_{32}	ω_{33}	ε_{34}	ϑ_{35}	ϖ_{36}	ϱ_{37}	ς_{38}	φ_{39}	″2x
′05x	$\leftharpoonup_{40}$	$\leftharpoondown_{41}$	$\rightharpoonup_{42}$	$\rightharpoondown_{43}$	$\triangleleft_{44}$	$\triangleright_{45}$	$\triangleright_{46}$	$\triangleleft_{47}$	
′06x	0_{48}	1_{49}	2_{50}	3_{51}	4_{52}	5_{53}	6_{54}	7_{55}	″3x
′07x	8_{56}	9_{57}	$.{}_{58}$	$,{}_{59}$	$<_{60}$	$/_{61}$	$>_{62}$	$\star_{63}$	
′10x	∂_{64}	A_{65}	B_{66}	C_{67}	D_{68}	E_{69}	F_{70}	G_{71}	″4x
′11x	H_{72}	I_{73}	J_{74}	K_{75}	L_{76}	M_{77}	N_{78}	O_{79}	
′12x	P_{80}	Q_{81}	R_{82}	S_{83}	T_{84}	U_{85}	V_{86}	W_{87}	″5x
′13x	X_{88}	Y_{89}	Z_{90}	$\flat_{91}$	$\natural_{92}$	$\sharp_{93}$	$\smile_{94}$	$\frown_{95}$	
′14x	ℓ_{96}	a_{97}	b_{98}	c_{99}	d_{100}	e_{101}	f_{102}	g_{103}	″6x
′15x	h_{104}	i_{105}	j_{106}	k_{107}	l_{108}	m_{109}	n_{110}	o_{111}	
′16x	p_{112}	q_{113}	r_{114}	s_{115}	t_{116}	u_{117}	v_{118}	w_{119}	″7x
′17x	x_{120}	y_{121}	z_{122}	$\imath_{123}$	$\jmath_{124}$	$\wp_{125}$	$\vec{}_{126}$	$\widehat{}_{127}$	
′20x	$_{128}$	$\varkappa_{129}$	$_{130}$	$_{131}$	0_{132}	1_{133}	2_{134}	3_{135}	″8x
′21x	4_{136}	5_{137}	6_{138}	7_{139}	8_{140}	9_{141}	A_{142}	B_{143}	
′22x	C_{144}	D_{145}	E_{146}	F_{147}	G_{148}	H_{149}	I_{150}	J_{151}	″9x
′23x	K_{152}	L_{153}	M_{154}	N_{155}	O_{156}	P_{157}	Q_{158}	R_{159}	
′24x	S_{160}	T_{161}	U_{162}	V_{163}	W_{164}	X_{165}	Y_{166}	Z_{167}	″Ax
′25x	a_{168}	b_{169}	c_{170}	d_{171}	e_{172}	f_{173}	g_{174}	h_{175}	
′26x	i_{176}	j_{177}	k_{178}	l_{179}	m_{180}	n_{181}	o_{182}	p_{183}	″Bx
′27x	q_{184}	r_{185}	s_{186}	t_{187}	u_{188}	v_{189}	w_{190}	x_{191}	
′30x	y_{192}	z_{193}	$\imath_{194}$	$\jmath_{195}$	A_{196}	B_{197}	C_{198}	D_{199}	″Cx
′31x	E_{200}	F_{201}	G_{202}	H_{203}	I_{204}	J_{205}	K_{206}	L_{207}	
′32x	M_{208}	N_{209}	O_{210}	P_{211}	Q_{212}	R_{213}	S_{214}	T_{215}	″Dx
′33x	U_{216}	V_{217}	W_{218}	X_{219}	Y_{220}	Z_{221}	a_{222}	b_{223}	
′34x	c_{224}	d_{225}	e_{226}	f_{227}	g_{228}	h_{229}	i_{230}	j_{231}	″Ex
′35x	k_{232}	l_{233}	m_{234}	n_{235}	o_{236}	p_{237}	q_{238}	r_{239}	
′36x	s_{240}	t_{241}	u_{242}	v_{243}	w_{244}	x_{245}	y_{246}	z_{247}	″Fx
′37x	$\imath_{248}$	$\jmath_{249}$	$_{250}$	$_{251}$	$_{252}$	$_{253}$	$_{254}$	$_{255}$	
	″8	″9	″A	″B	″C	″D	″E	″F	

Serif Math LettersA

	′0	′1	′2	′3	′4	′5	′6	′7	
′00x	Γ0	Δ1	Θ2	Λ3	Ξ4	Π5	Σ6	Υ7	″0x
′01x	Φ8	Ψ9	Ω10	α11	β12	γ13	δ14	ϵ15	
′02x	ζ16	η17	θ18	ι19	κ20	λ21	μ22	ν23	″1x
′03x	ξ24	π25	ρ26	σ27	τ28	υ29	ϕ30	χ31	
′04x	ψ32	ω33	ε34	ϑ35	ϖ36	ϱ37	ς38	φ39	″2x
′05x	40	41	42	𝟘43	𝟙44	𝟚45	𝟛46	𝟜47	
′06x	𝟝48	𝟞49	𝟟50	𝟠51	𝟡52	53	∉54	∌55	″3x
′07x	∉56	∌57	:=58	=:59	≠60	=61	{62	}63	
′10x	∂64	𝔄65	𝔅66	ℭ67	𝔇68	𝔈69	𝔉70	𝔊71	″4x
′11x	ℌ72	ℑ73	𝔍74	ℜ75	𝔏76	𝔐77	𝔑78	𝔒79	
′12x	𝔓80	𝔔81	ℜ82	𝔖83	𝔗84	𝔘85	𝔙86	𝔚87	″5x
′13x	𝔛88	𝔜89	ℨ90	ℏ91	ℎ92	𝜆93	𝜁94	Å95	
′14x	Ɛ96	𝔞97	𝔟98	𝔠99	𝔡100	𝔢101	𝔣102	𝔤103	″6x
′15x	𝔥104	𝔦105	𝔧106	𝔨107	𝔩108	𝔪109	𝔫110	𝔬111	
′16x	𝔭112	𝔮113	𝔯114	𝔰115	𝔱116	𝔲117	𝔳118	𝔴119	″7x
′17x	𝔵120	𝔶121	𝔷122	𝔱123	𝔦124	𝕀125	𝕁126	̂127	
′20x	128	𝜘129	130	131	𝔸132	𝔹133	ℂ134	𝔻135	″8x
′21x	𝔼136	𝔽137	𝔾138	ℍ139	𝕀140	𝕁141	𝕂142	𝕃143	
′22x	𝕄144	ℕ145	𝕆146	ℙ147	ℚ148	ℝ149	𝕊150	𝕋151	″9x
′23x	𝕌152	𝕍153	𝕎154	𝕏155	𝕐156	ℤ157	𝕒158	𝕓159	
′24x	𝕔160	𝕕161	𝕖162	𝕗163	𝕘164	𝕙165	𝕚166	𝕛167	″Ax
′25x	𝕜168	𝕝169	𝕞170	𝕟171	𝕠172	𝕡173	𝕢174	𝕣175	
′26x	𝕤176	𝕥177	𝕦178	𝕧179	𝕨180	𝕩181	𝕪182	𝕫183	″Bx
′27x	𝕝184	𝕁185	𝕝186	𝕁187	𝑔188	𝑦189	190	191	
′30x	192	𝔸193	𝔹194	ℂ195	𝔻196	𝔼197	𝔽198	𝔾199	″Cx
′31x	ℍ200	𝕀201	𝕁202	𝕂203	𝕃204	𝕄205	ℕ206	𝕆207	
′32x	ℙ208	ℚ209	ℝ210	𝕊211	𝕋212	𝕌213	𝕍214	𝕎215	″Dx
′33x	𝕏216	𝕐217	ℤ218	𝛤219	𝛱220	𝛶221	𝜛222	223	
′34x	224	𝕒225	𝕓226	𝕔227	𝕕228	𝕖229	𝕗230	𝕘231	″Ex
′35x	𝕙232	𝕚233	𝕛234	𝕜235	𝕝236	𝕞237	𝕟238	𝕠239	
′36x	𝕡240	𝕢241	𝕣242	𝕤243	𝕥244	𝕦245	𝕧246	𝕨247	″Fx
′37x	𝕩248	𝕪249	𝕫250	𝜆251	252	253	254	255	
	″8	″9	″A	″B	″C	″D	″E	″F	